

# Customer Churn Prediction Summary and Insight

## Summary:

The project aimed to develop a machine learning solution to predict customer churn using a provided classification dataset. The end-to-end process included data analysis, preprocessing, model training, and evaluation.

## Key Steps:

- Exploratory Data Analysis (EDA): Revealed a class imbalance, with churners comprising approximately 18.8% of the dataset, and identified features with significant missing data.
- Data Preprocessing: Columns with more than 90% missing values were dropped. Remaining missing values were imputed (median for numerical, mode for categorical). Categorical features were one-hot encoded, and numerical features were scaled.
- Handling Imbalance: SMOTE (Synthetic Minority Over-sampling Technique) was used to balance the training dataset.
- Model Training: Several classifiers were evaluated, including Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost.
- Model Selection: The tuned XGBoost model performed the best overall.

## Performance of Tuned XGBoost:

- Accuracy: 0.81
- Precision (Churn=1): 0.61
- Recall (Churn=1): 0.20
- F1-Score (Churn=1): 0.30
- ROC AUC Score: 0.70

Feature importance analysis from the XGBoost model identified key predictors of churn, supporting future targeted retention strategies.

## Insights

### 1. Model Performance Trade-offs:

The tuned XGBoost model shows solid accuracy and precision, especially useful for targeting customers flagged as churn risks.

However, the low recall (0.20) indicates the model misses 80% of actual churners - a critical drawback for intervention planning.

The F1-score (0.30) reflects this imbalance, suggesting a need to optimize for recall even if precision drops slightly.

### 2. Impact of Class Imbalance:

Despite using SMOTE, the model struggled with identifying churners, indicating that either more advanced imbalance techniques are needed or the available features are not sufficiently discriminative.

### 3. Data Quality & Feature Relevance:

Several features had to be dropped due to missing data (>90%), signaling potential issues in data collection or their business relevance.

Top features (e.g., cat\_\_Var207\_KASS, cat\_\_Var217\_smXZ, num\_\_Var189) strongly influenced predictions. Understanding these from a domain/business perspective is essential to extract actionable insights.

### 4. Actionable Recommendations:

- Enhance Recall: Explore alternative techniques like cost-sensitive learning, ensemble strategies focused on the minority class, or better feature engineering.
- Business Metric Alignment: If reducing missed churners is critical, prioritize recall over precision.
- Feature Interpretation: Investigate the business meaning of the top predictive features to support strategic interventions and customer retention initiatives.